

TABLE 59 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	1.02	1.19	1.39	1.62	1.89	2.21	2.58	3.04	16.9%
Japan	0.48	0.56	0.66	0.77	0.90	1.06	1.25	1.47	17.6%
India	0.06	0.07	0.08	0.10	0.11	0.13	0.15	0.18	16.8%
Rest of Asia Pacific	0.86	1.00	1.17	1.36	1.58	1.84	2.15	2.53	16.7%
Total	2.42	2.83	3.29	3.84	4.48	5.23	6.13	7.22	17.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 60 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SBL, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.26	0.29	0.32	0.35	0.39	0.43	0.48	0.53	10.8%
Rest of Latin America	0.29	0.32	0.36	0.39	0.44	0.48	0.53	0.60	10.7%
Total	0.55	0.61	0.68	0.75	0.82	0.91	1.01	1.13	10.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.4 NANOPORE SEQUENCING

8.4.1 ONLY SEQUENCING TECHNOLOGY WITH DIRECT RNA SEQUENCING CAPABILITIES

Introduced by Oxford Nanopore Technologies, nanopore sequencing technology (which is considered the fourth generation of sequencing) has emerged as a competitive, portable technology.

The technology works through the detection of unique electrical signals of different molecules as they pass across the nanopore with a semiconductor-based electronic detection system. This technology makes for a high throughput, cost-effective sequencing solution. The technology's core/heart consists of the biological nanopore, a protein pore embedded in a membrane, while the brains of the technology lie in the electronics of a semiconductor integrated circuit and proprietary chemistries. The electronic sensor embedded in the chip enables automatic membrane assembly and nanopore insertion while allowing for active control of individual sensors on the circuit. Various types of sequencing chemistries can be paired with nanopore and electronic sensor technology, which enables high-throughput and high-accuracy sequencing. However, the low accuracy of nanopore sequencing as compared to other well-established techniques is one of the major factors restraining the growth of this market segment.

In mid-2014, Oxford Nanopore Technologies launched MinION, which was the only nanopore sequencer commercially available in the market until 2015. This sequencer is portable, weighs just 100 gm, and can sequence DNA and RNA of up to ~200kb in length. Due to its portable nature, the MinION sequencer can

be used for real-time sequencing. From January to July 2019, the company launched GridION, Flongle sequencer, and PromethION 24 and PromethION 48 (on the e-Commerce platform of the company to make them available globally). Also, in May 2018, the company upgraded the MinKnow software that drives the MinION, GridION, and PromethION sequencing devices.

TABLE 61 MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	6.47	7.47	8.62	9.95	11.49	13.30	15.46	18.03	15.9%
Europe	3.49	4.12	4.86	5.72	6.74	7.96	9.43	11.21	18.2%
Asia Pacific	0.96	1.18	1.44	1.76	2.14	2.62	3.20	3.92	22.2%
Latin America	0.22	0.26	0.30	0.35	0.41	0.47	0.55	0.64	16.5%
Middle East and Africa	0.15	0.18	0.20	0.24	0.27	0.31	0.36	0.41	15.2%
Total	11.29	13.20	15.42	18.01	21.05	24.67	29.00	34.21	17.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 62 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	5.87	6.78	7.82	9.03	10.43	12.09	14.05	16.40	16.0%
Canada	0.61	0.70	0.80	0.92	1.06	1.22	1.41	1.63	15.4%
Total	6.47	7.47	8.62	9.95	11.49	13.30	15.46	18.03	15.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 63 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	0.76	0.89	1.04	1.23	1.44	1.70	2.00	2.38	17.9%
UK	0.68	0.80	0.95	1.12	1.33	1.58	1.88	2.24	18.8%
France	0.53	0.63	0.74	0.88	1.03	1.22	1.45	1.72	18.3%
Italy	0.15	0.18	0.21	0.25	0.29	0.34	0.40	0.48	18.1%
Spain	0.12	0.14	0.16	0.19	0.22	0.26	0.31	0.37	18.0%
Rest of Europe	1.26	1.49	1.75	2.06	2.43	2.86	3.39	4.02	18.1%
Total	3.49	4.12	4.86	5.72	6.74	7.96	9.43	11.21	18.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 64 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	0.41	0.50	0.61	0.74	0.90	1.09	1.33	1.63	21.9%
Japan	0.19	0.23	0.29	0.35	0.43	0.53	0.65	0.80	22.8%
India	0.02	0.03	0.04	0.04	0.05	0.07	0.08	0.10	22.2%
Rest of Asia Pacific	0.34	0.42	0.51	0.63	0.76	0.93	1.14	1.40	22.2%
Total	0.96	1.18	1.44	1.76	2.14	2.62	3.20	3.92	22.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 65 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR NANOPORE SEQUENCING, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.10	0.12	0.14	0.16	0.19	0.22	0.26	0.30	16.5%
Rest of Latin America	0.12	0.14	0.16	0.19	0.22	0.25	0.29	0.34	16.6%
Total	0.22	0.26	0.30	0.35	0.41	0.47	0.55	0.64	16.5%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

8.5 OTHER TECHNOLOGIES

Thermo Fisher's ion semiconductor sequencing technology-based platforms include Ion Torrent PGM, Ion Proton, and the newly launched GeneStudio S5 Series. Ion semiconductor sequencing involves the direct translation of genetic codes of sequence (DNA bases) into digital information onto a semiconductor chip. The technology translates chemically encoded information (A, C, G, T) into digital information (0, 1) on a semiconductor chip. The company focuses on continuous enhancements in the technology to widen its scope of applications and increase its precision of sequencing.

The launch of new and advanced sequencers by Thermo Fisher and the advantages offered by ion semiconductor technology platforms are expected to drive market growth in the near future. For instance, in March 2022, the company launched the Ion Torrent Genexus Dx System. This fully validated system enables users to perform both diagnostic testing and clinical research on a single instrument. The company also provides complete NGS workflow for diagnostic solutions for comprehensive genomic profiling. The high adoption of ion semiconductor technology can be attributed to its features, including simpler workflow, time- as well as cost-efficiency, and scalability. This technology is gaining popularity in microbial profiling, thus contributing to the overall segment growth.

TABLE 66 MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%
Europe	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%
Asia Pacific	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%
Latin America	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%
Middle East and Africa	1.10	1.19	1.29	1.39	1.50	1.63	1.76	1.91	8.1%
Total	80.97	91.63	103.64	117.32	132.92	151.11	172.40	197.53	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 67 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	41.85	47.25	53.33	60.24	68.11	77.27	87.96	100.57	13.5%
Canada	4.33	4.85	5.42	6.07	6.80	7.65	8.62	9.77	12.5%
Total	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis